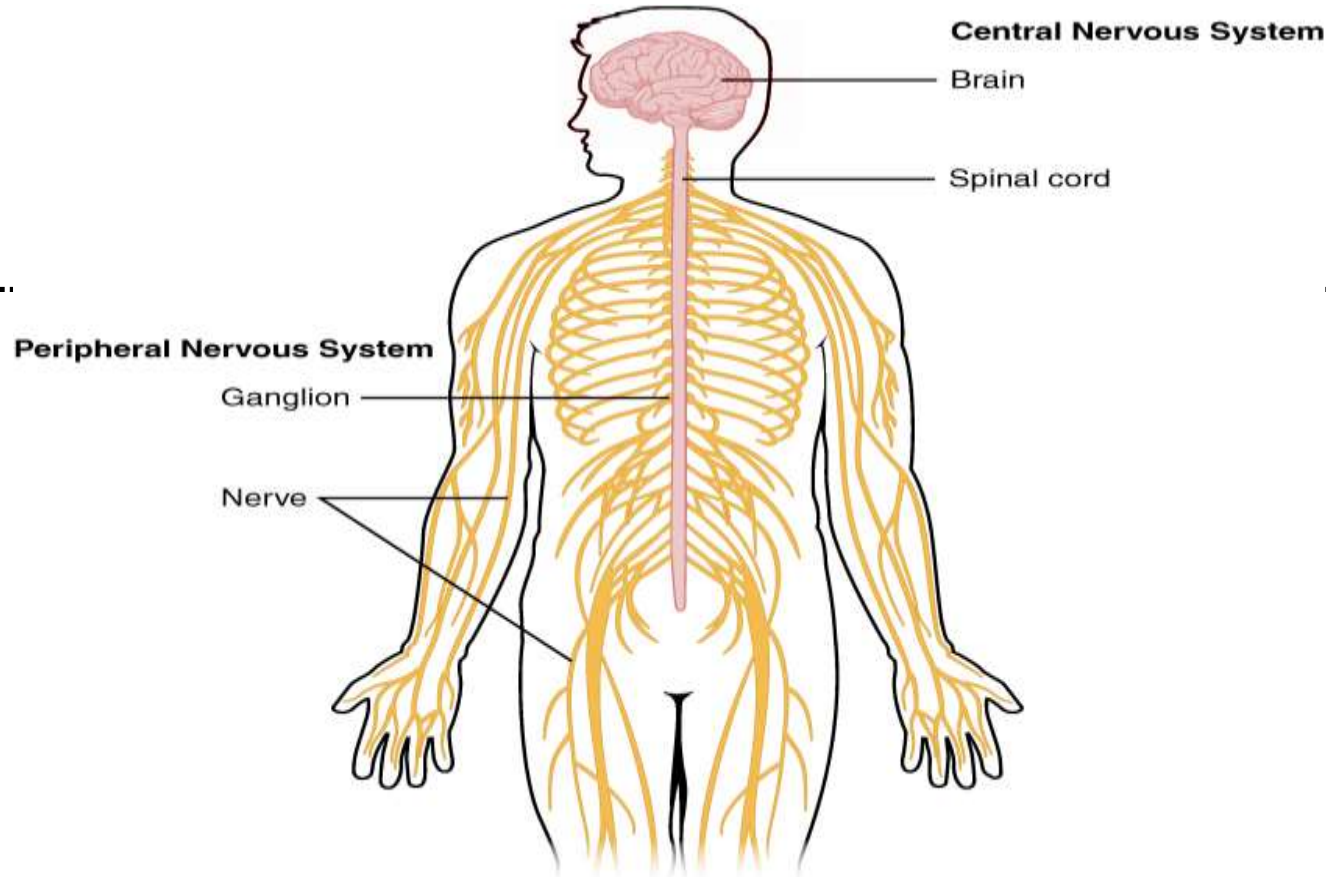
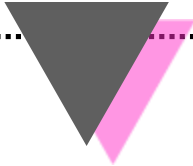
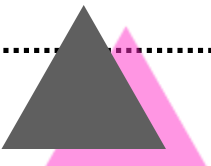


Human Physiology: The Nervous System



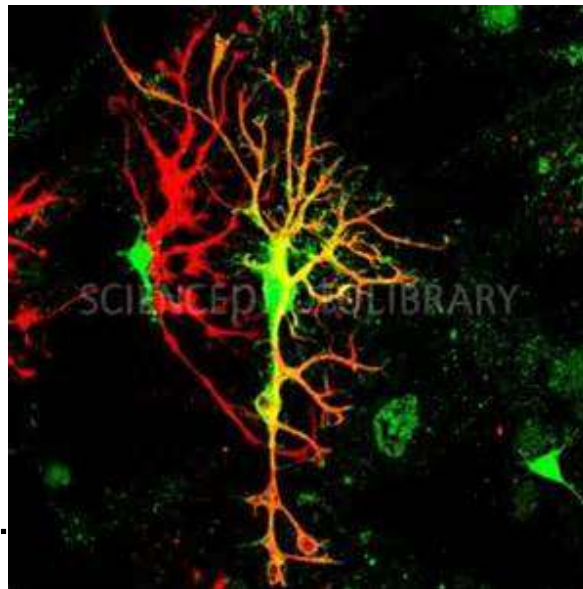


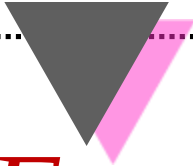
Objectives and Questions:

1. I can describe the functions of the nervous system
 2. I can describe the parts of a neuron cell and identify how they transmit electrochemical impulses.
 3. I can compare and contrast the central and peripheral nervous systems
 4. I can identify and explain different areas of the brain and their functions.
 5. I can explain how the nervous system passes information between the external environment and the many parts of the body.
- 

The Nervous system is:

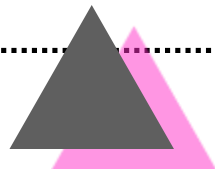
A physically connected network of cells, tissues and organs that allow us to communicate with and react to the environment and perform life activities.





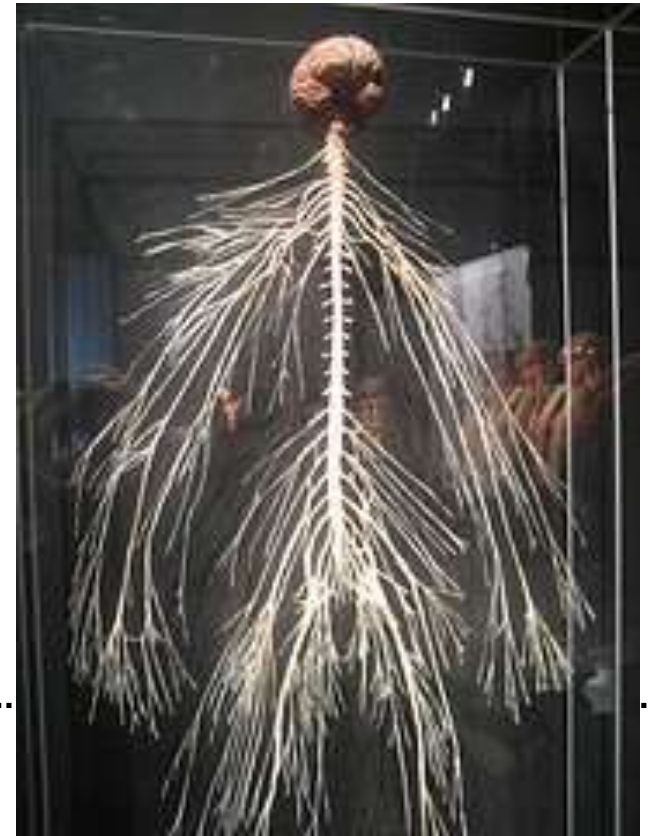
Cool Nervous System Facts:

- As a fetus in the uterus, neurons develop at the rate of 250,000 per minute.
- We have about 100 billion nerve cells in our brain by adulthood (over 600 miles worth) – yet we typically use around 4% at any given time
- Neurons are our largest cells
- Messages transmit at speeds up of to 180 m/s (268 mile/h or 180 m/s)



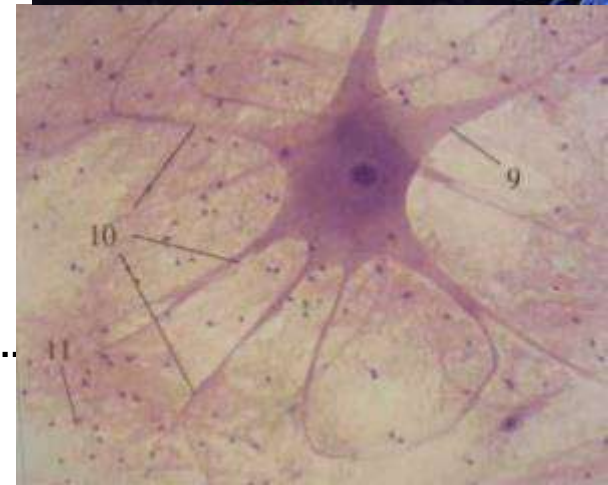
FOUR PRIMARY FUNCTIONS OF NERVOUS SYSTEM

1. **Sensing the world**
 - **Vision, Hearing, Smell, Taste, Touch**
2. **Transmitting information**
3. **Processing information**
4. **Producing a response**



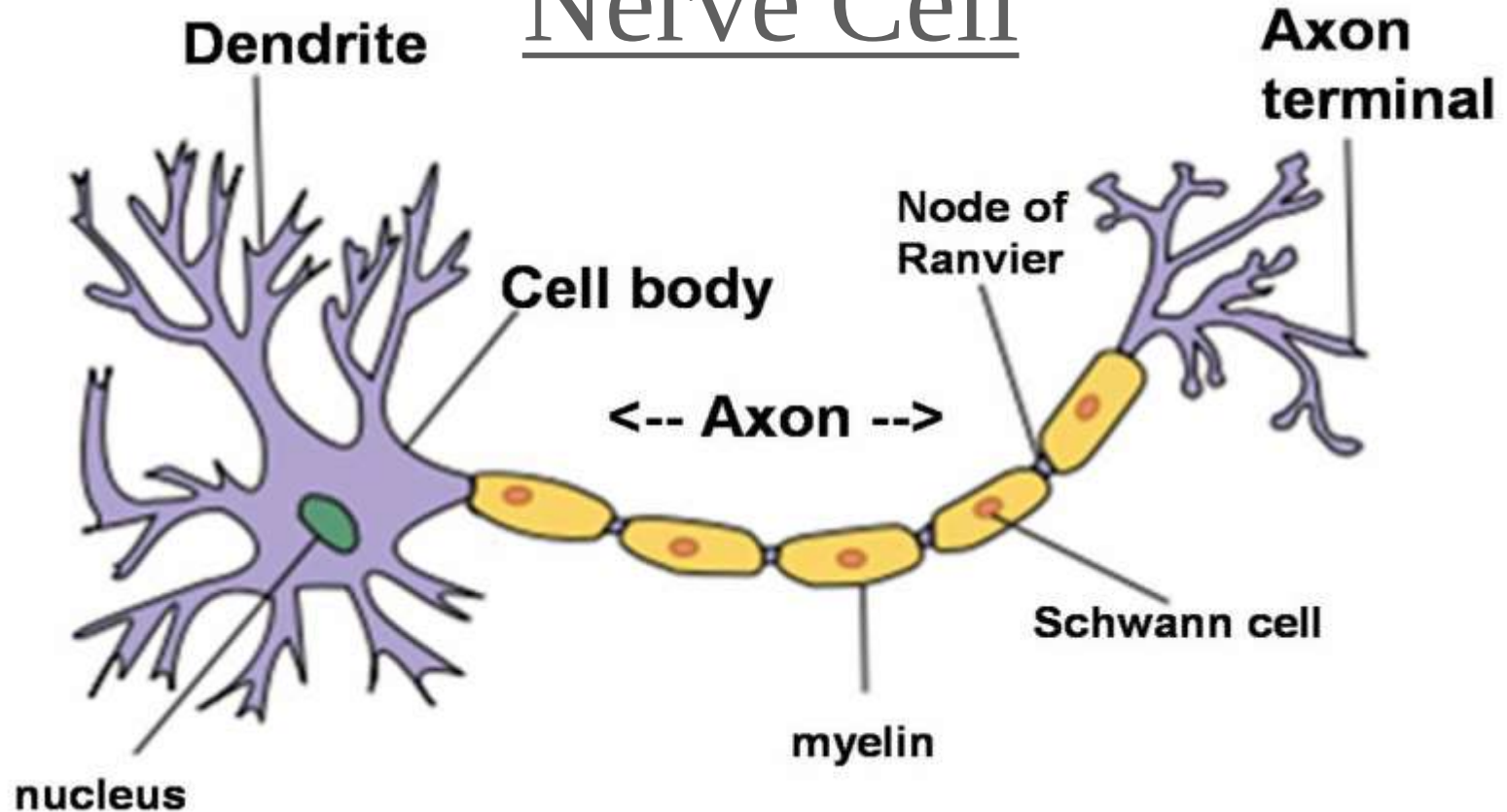
Neurons

- The functional unit of the nervous system is the nerve cell, the **NEURON**
 - They send electrochemical messages around the body
- *Glial cells* provide support and protection for neurons



Parts of a neuron

Nerve Cell



Parts of nerve cell

Dendrites : receive chemical signals from neighboring cells.

Cell Body: contains the nucleus & organelles

Axon: long extension that carries electrical messages away from the body to the terminal axons

Terminal Axons: passes the signal to the next cell.

Myelin sheath: Protective covering for axon

Types of Neurons

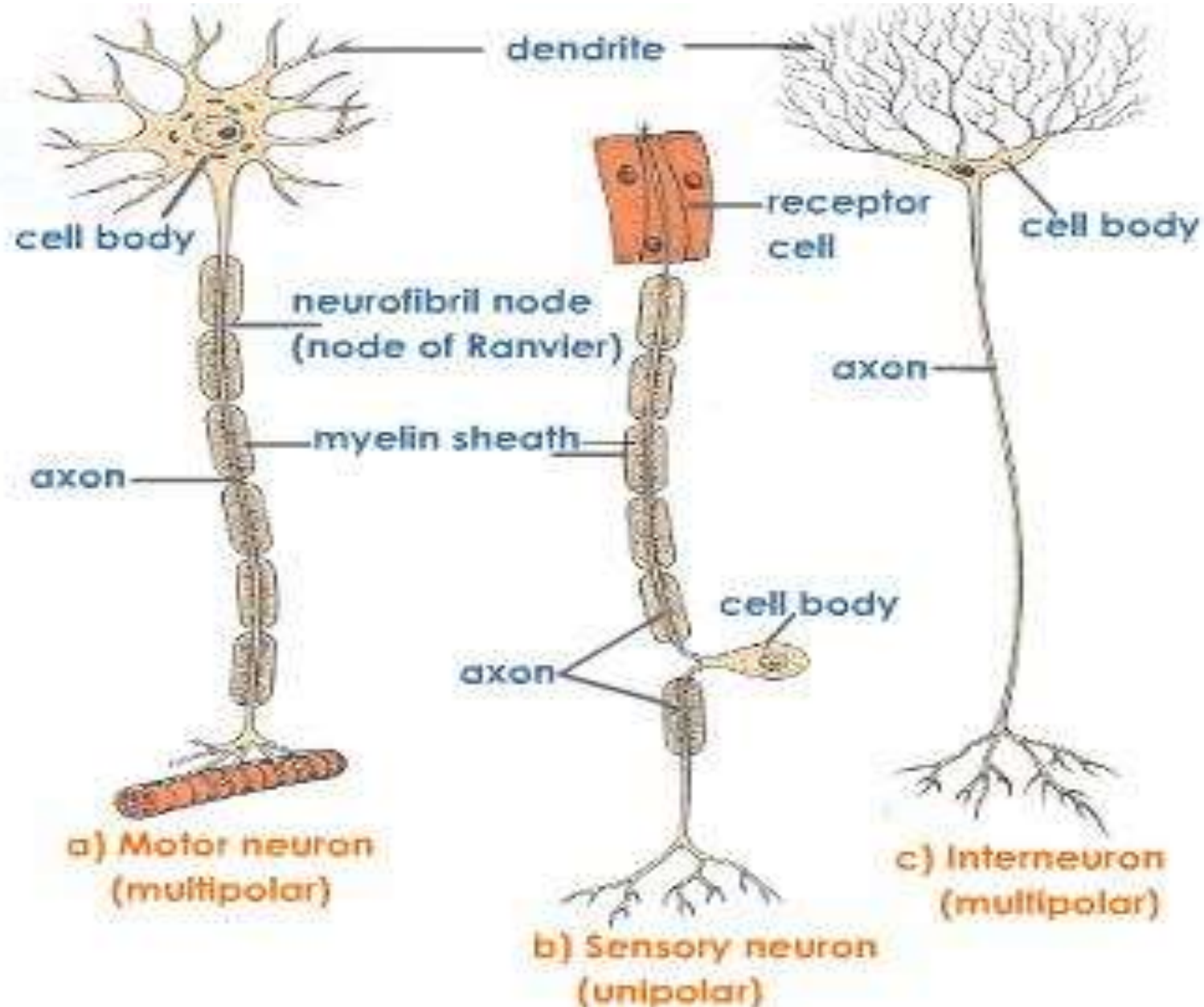
3 main types of neurons:

Sensory neuron = detect stimuli

Interneurons = relay signals between neurons, for example they relay sensory signals to brain then return message back to motor neurons.

Motor neuron = pass message from brain or spinal cord to rest of body for muscle response.

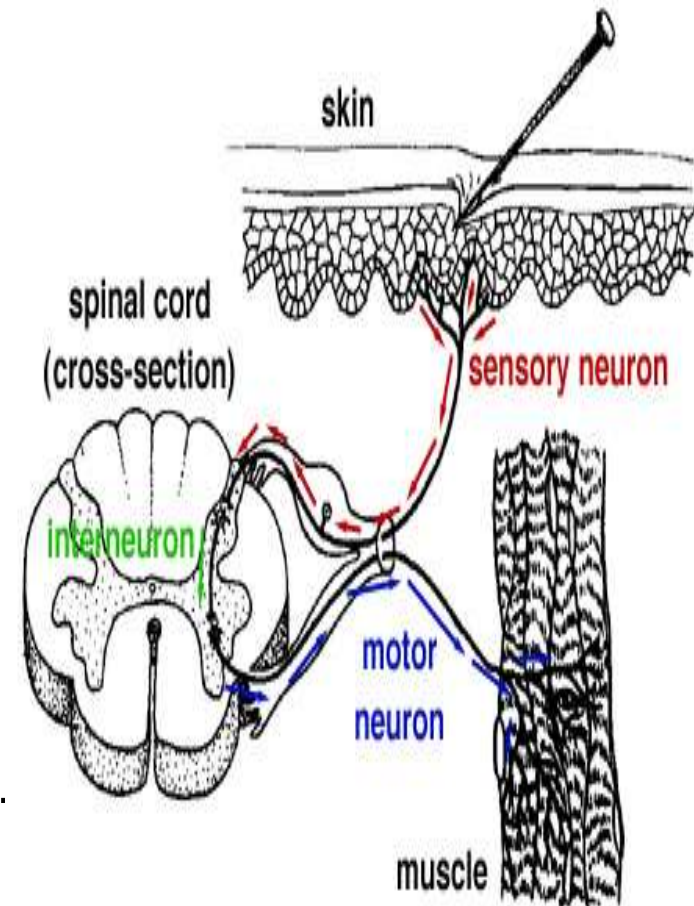
This coordinated pathway is known as the
REFLEX ARC

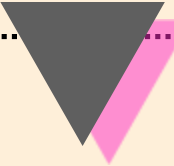


Reflex Arc

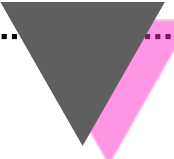
“ what happens when you step on a nail”?

- *Reflexes are automatic unplanned response to a stimulus.*
- The Stimulus (nail) is received by the **sensory neurons** in the foot
- This info. travels to the spine, where the **interneuron** is triggered
- The **interneuron** transmits signal to brain (through the spinal cord)and carries message back and stimulates the **motor** neuron, to move the foot





Somatic Sensation



What is somatic sensation?

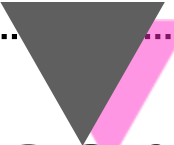
How do we detect changes in our environment?

How strong is the stimulus?

How long does the stimulus last?

Where is the stimulus?

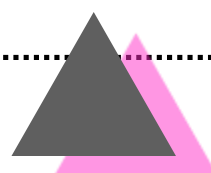
Where do we 'feel' sensation?



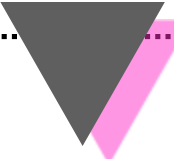
Introduction to Sensory System

Sensation

is a conscious or sub-conscious awareness of an external or internal stimulus



What is somatic sensation?



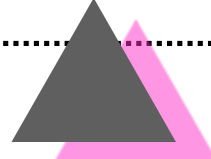
The Senses

- General Senses

- (a) **Somatic** Sensation from body
 - i.e., Tactile (touch, pressure, vibration)
 - Thermal (warm, cold)
 - Pain
 - Proprioception
- (b) **Visceral** Sensation from internal organs

- Special Senses

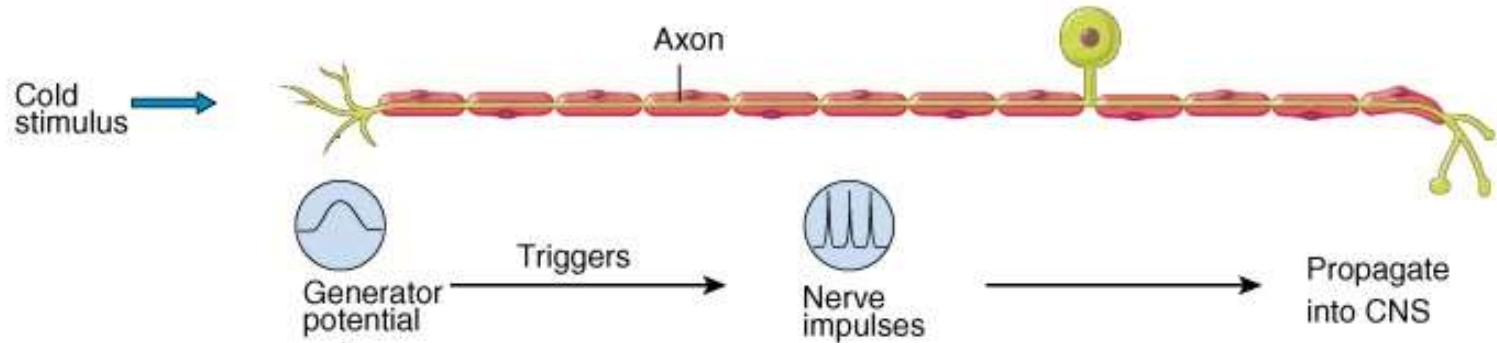
(smell, taste, vision, hearing)



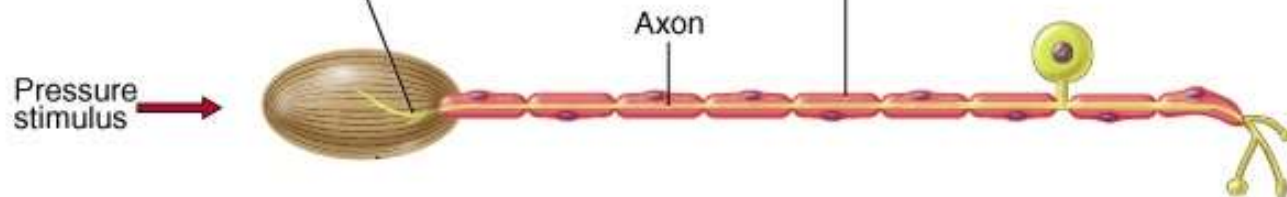
What is somatic sensation?

Sensory Receptors of 1st Order Neurones

**Free
Nerve
Ending**



**Encap-
sulated
Nerve
Ending**

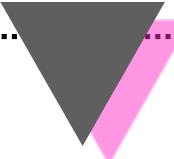


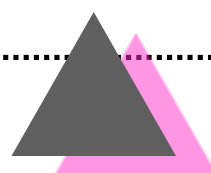
**With
Specialised
Cell**



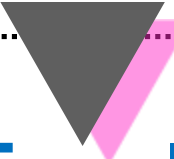
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What is somatic sensation?

- 
- **stimulus modalities** includes light, touch, temperature, chemical changes (e.g., taste) etc....
 - **qualities**: a subdivision of modality, e.g. taste can be divided into sweet, sour, salt etc
 - Sensory receptors are **modality specific**



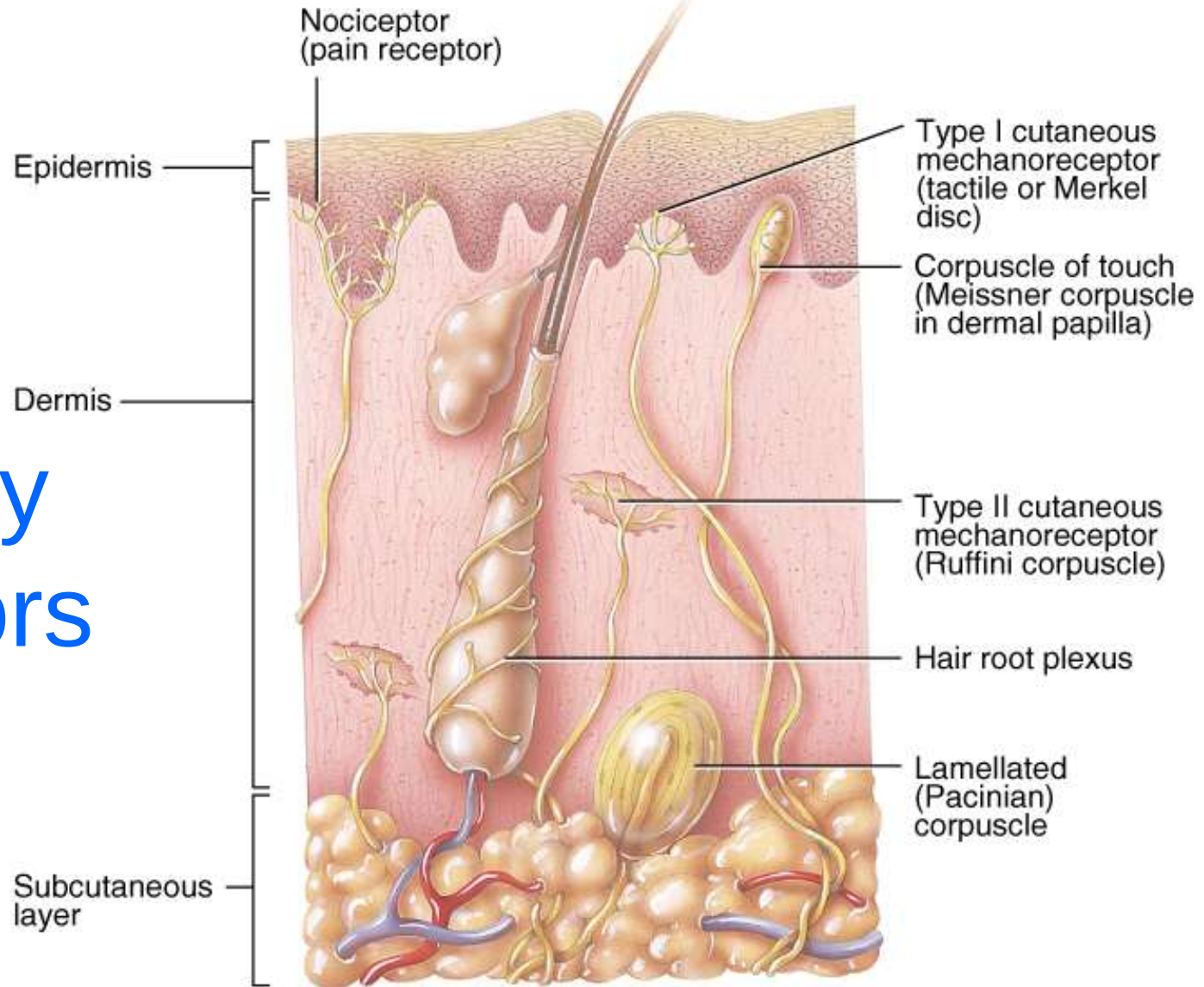
What is somatic sensation?



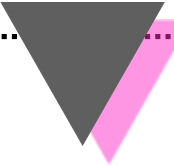
CNS Distinguishes 4 Stimulus Properties

- Modality (nature) of stimulus
 - Type of receptor
- Location
- Intensity
- Duration

Sensory Receptors of the Skin



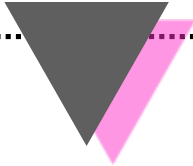
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Mechanoreceptors

which comprise both **exteroceptors** and **proprioceptors**, are activated following **physical deformation** due to touch, pressure, stretch, or vibration of the skin, muscles, tendons, ligaments, and joint capsules, in which they reside.

A mechanoreceptor may be **nonencapsulated** or **encapsulated**



Nonencapsulated mechanoreceptors

Nonencapsulated mechanoreceptors are **slowly adapting** and include *free nerve endings and some tactile receptors*:

Free nerve endings are present in the **epidermis, dermis, cornea, dental pulp, mucous membranes** of the **oral and nasal** cavities and of the **respiratory, gastrointestinal, and urinary tracts, muscles, tendons, ligaments, joint capsules, and bones**.

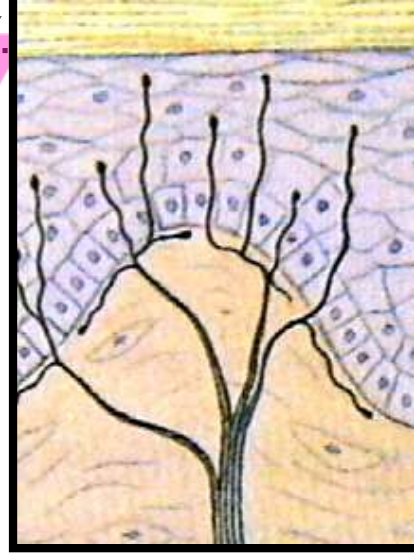
Touch Receptors (Tactile receptors)

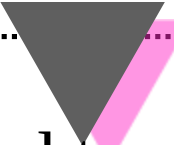
Free or encapsulated dendritic endings

In **skin** and **deep organs**, e.g.: Pacinian corpuscles

When stimulated, it opens mechanically gated ion channel

the type have **rapid adaptation** $\Rightarrow$
receptor type?



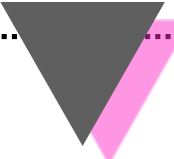


Tactile receptors (Merkel's discs), respond to discriminative touch stimuli that facilitate the distinguishing of texture, shape, and edges of objects.

Encapsulated mechanoreceptors : include:

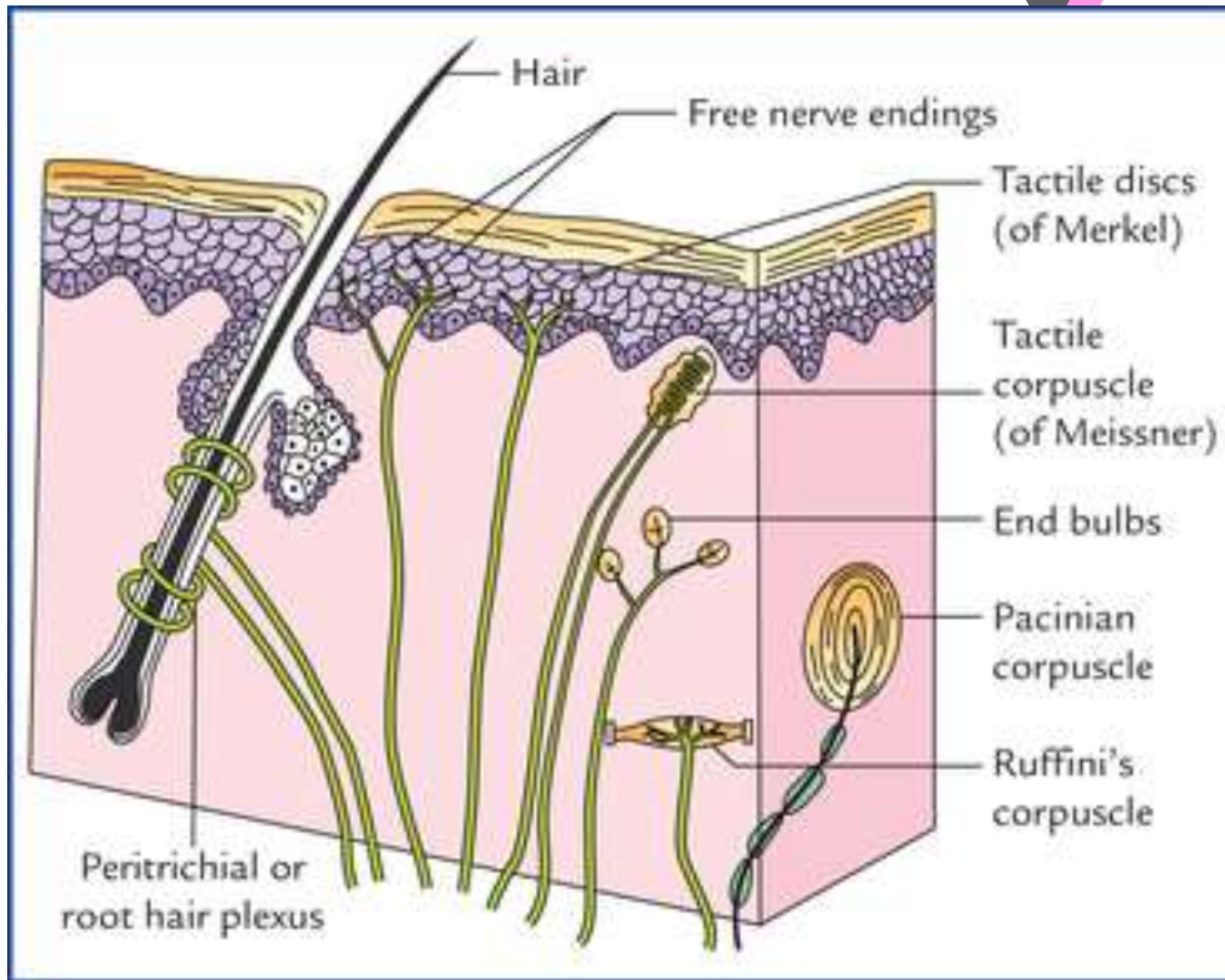
1- Meissner's corpuscles They are sensitive to two-point tactile (fine) discrimination, and are thus of great importance to the visually impaired by permitting them to be able to **read Braille**.

2- Pacinian corpuscles the **largest** of the mechanoreceptors. They are believed to respond to **pressure** and **vibratory** stimuli, including tickling sensations.



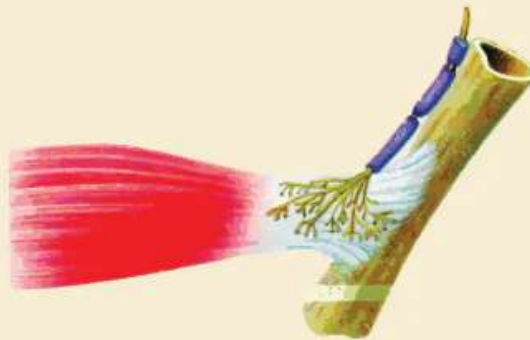
3- Ruffini's corpuscles respond to **stretching** of the collagen bundles in the skin or joint capsules and may provide **proprioceptive** information.

4- Muscle spindles and Golgi tendon organs The muscle spindles and GTOs detect **sensory input** from the **skeletal muscle** and transmit it to the spinal cord where it plays an important role in **reflex** activity and motor control involving the cerebellum

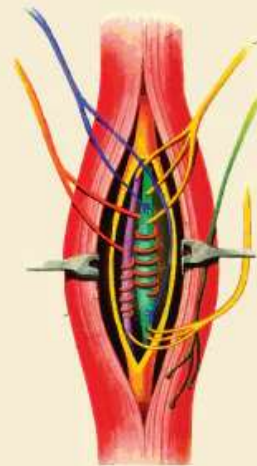


Muscle spindles and Golgi tendon organs

Golgi Tendon Organ VS Muscle Spindle

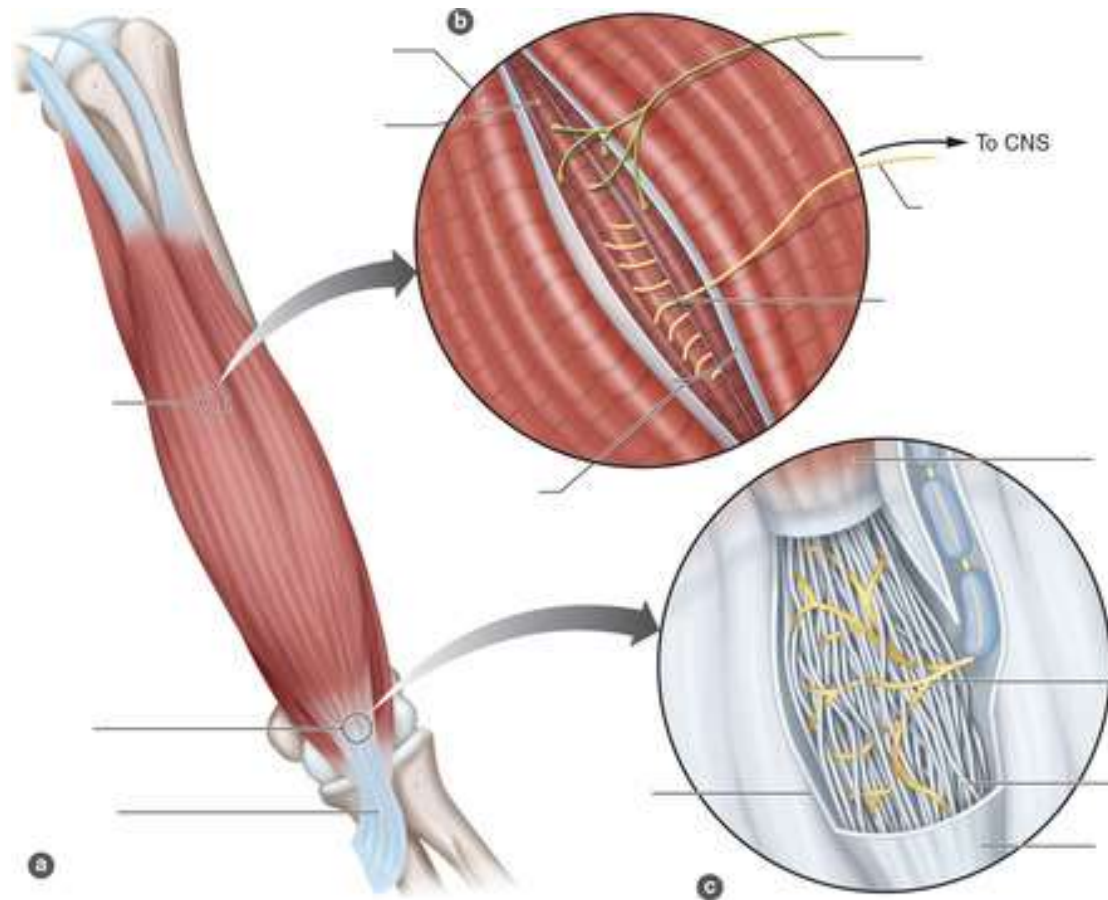


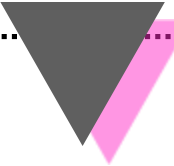
Golgi Tendon Organ



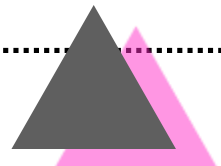
Muscle Spindle

Muscle spindles and Golgi tendon organs





Temperature Receptors

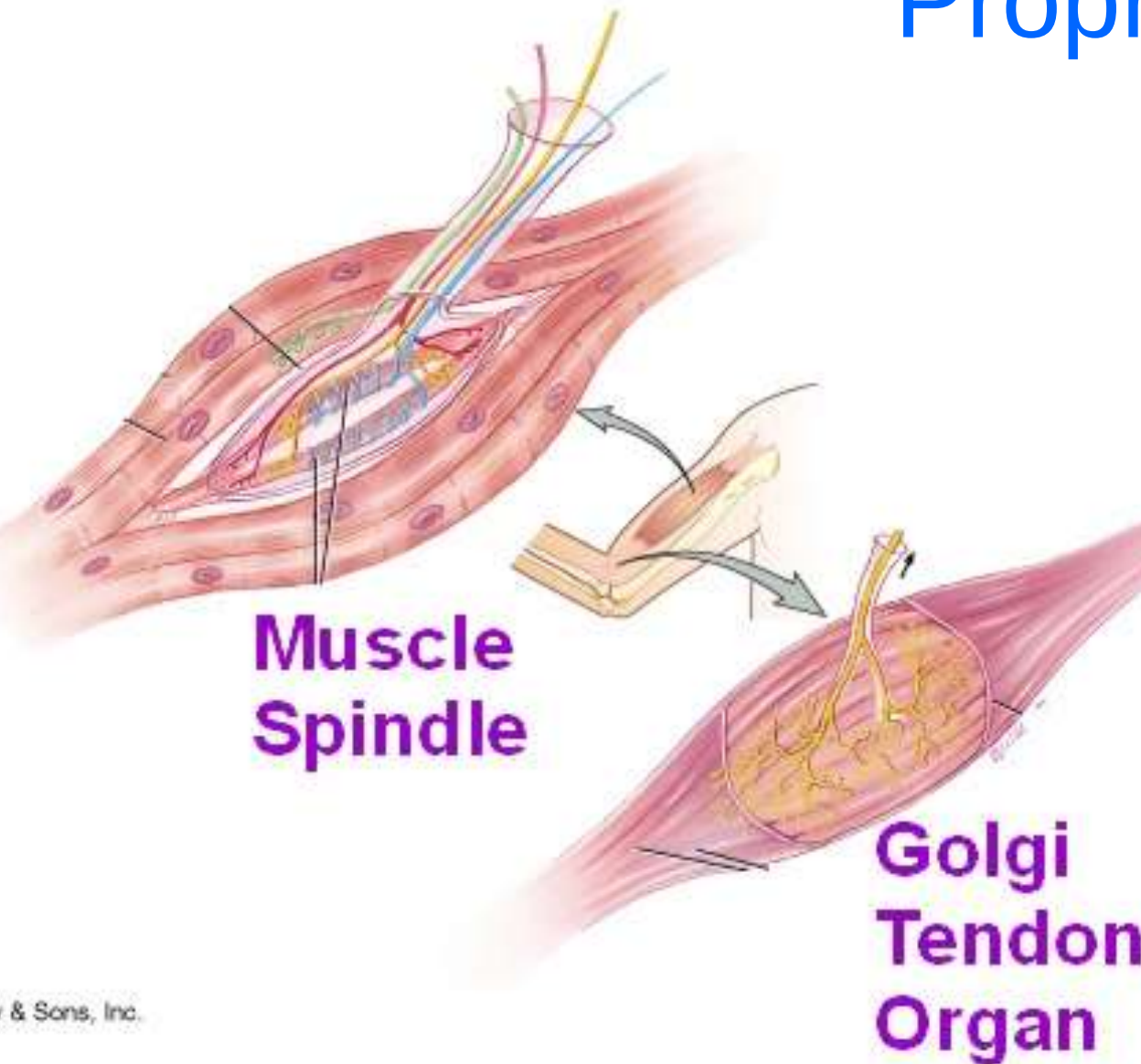
- thermoceptors
 - Free dendritic endings in hypodermis
 - Function in thermoregulation
 - **Cold receptors** (< body temp.)
 - **Warm receptors** (> body temp.)
 - Adaptation only between 20 and 40°C?
 - **Temperature-sensitive Nociceptors** activated is sensitive to excessive heat or cold.
- 

Nociceptors

- **Pain sensitive receptors.**
- **Free dendritic endings**
- **Activation by strong, noxious stimuli – *Function: transmitting pain***
- **3 categories:**
 - **Mechanical**
 - **Thermal**
 - **Chemical**
- **Polymodal nociceptors** (mechanical, thermal, or chemical painful stimuli).

Proprioceptors

- Sensory receptors of muscles and joints providing information on body position, i.e., proprioception





What is somatic sensation?

**How do we detect changes in
our environment?**

How strong is the stimulus?

How long does the stimulus last?

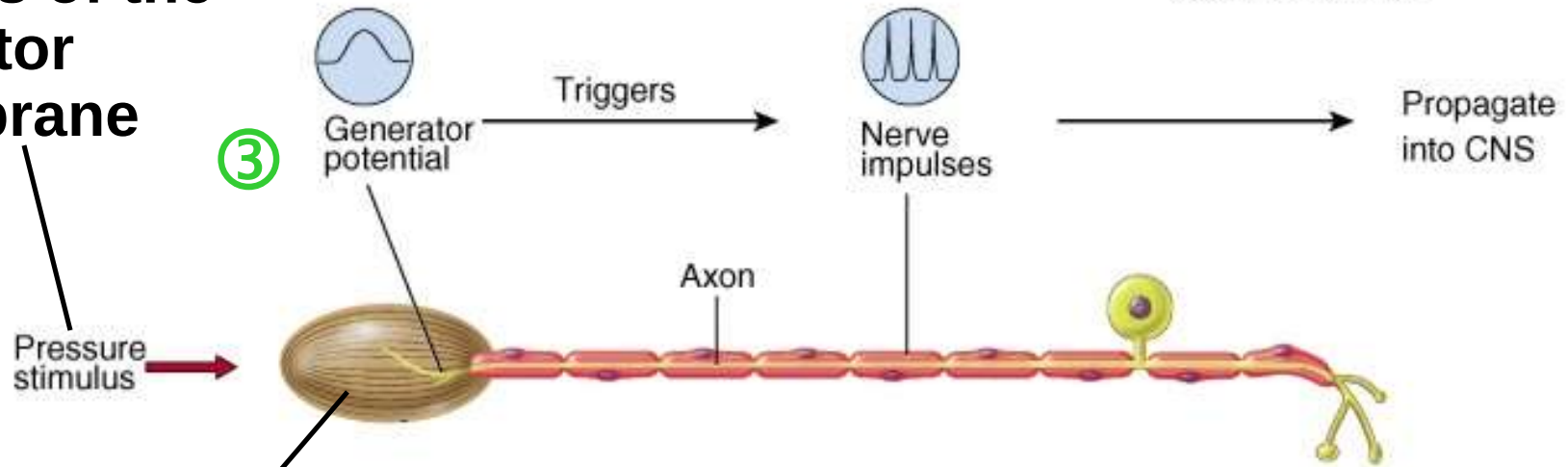
Where is the stimulus?

Where do we 'feel' sensation?

① Stimulus evokes change in the permeability to ions of the receptor membrane

② Receptor Potential (movement of ion across membrane)

④ Action Potentials



Sensory Transduction



What is somatic sensation?

How do we detect changes in our environment?

How strong is the stimulus?

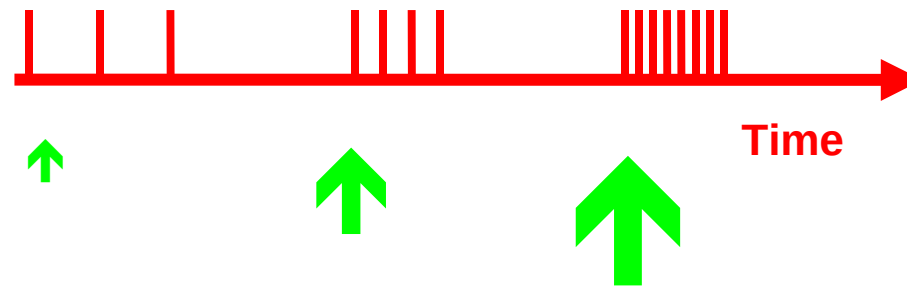
How long does the stimulus last?

Where is the stimulus?

Where do we 'feel' sensation?

Action
Potentials in
Afferent
Neurone

Stimulus
Strength



- **Strength (intensity):** determined by rate of action potential stimulus (**frequency coding**)
- A stronger stimulus also activates neighbouring cells (but to a lesser degree)

How Strong Is
the Stimulus?



What is somatic sensation?

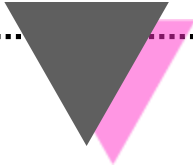
How do we detect changes in our environment?

How strong is the stimulus?

How long does the stimulus last?

Where is the stimulus?

Where do we 'feel' sensation?

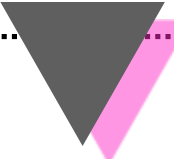


Duration is coded by duration of APs in sensory neurons

Sustained stimulation leads to **adaptation:**

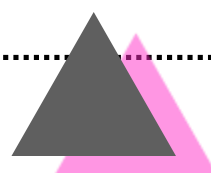
According to this we have:

- **Tonic receptors**
- **Phasic receptors**



Adaptation

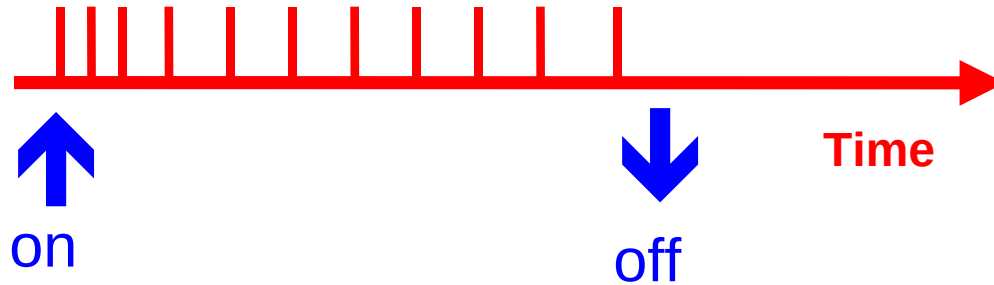
- Slowly adapting (**tonic**) receptors may keep firing as long as the stimulus lasts e.g., **joint receptors** and **baroreceptors**
- Rapidly adapting (**phasic**) receptors respond maximally and briefly to a stimulus e.g., **touch** receptors



How Long Does the Stimulus Last?

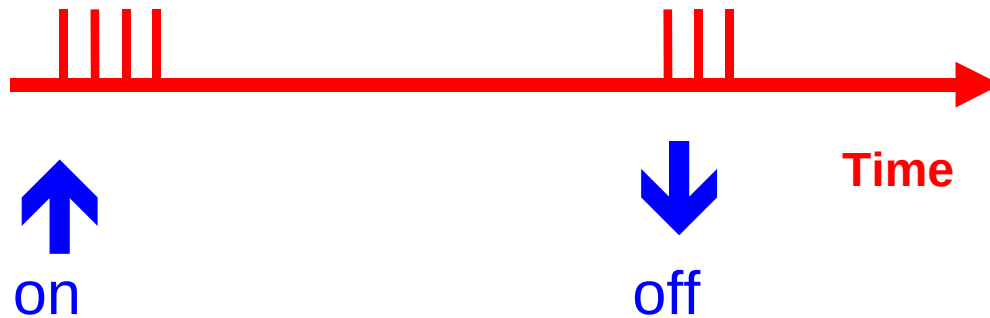
**Slowly
Adapting
Receptor**

Stimulus

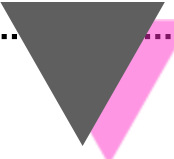


**Rapidly
Adapting
Receptor**

Stimulus



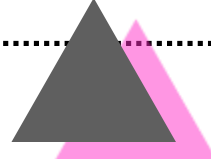
How Long Does the
Stimulus Last?



Tonic Receptors

- Slow or no adaptation
- Continuous signal transmission for duration of stimulus
- Monitoring of parameters that must be continually evaluated, e.g.: **baroreceptors**

Phasic Receptors

- Rapid adaptation
 - Cease firing if strength of a continuous stimulus remains constant
 - Allow body to ignore constant unimportant information, e.g.: **Smell**
- 



What is somatic sensation?

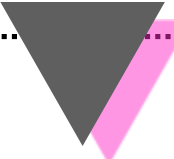
How do we detect changes in our environment?

How strong is the stimulus?

How long does the stimulus last?

Where is the stimulus?

Where do we 'feel' sensation?

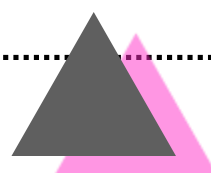


Sensory Acuity

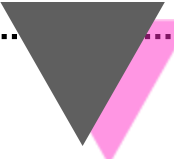
Acuity : the precision by which a stimulus can be located (or the ability to be acutely aware of the detail)

Determined by

1. lateral inhibition in the CNS
2. two point discrimination
3. synaptic convergence and divergence



Where is the
Stimulus?



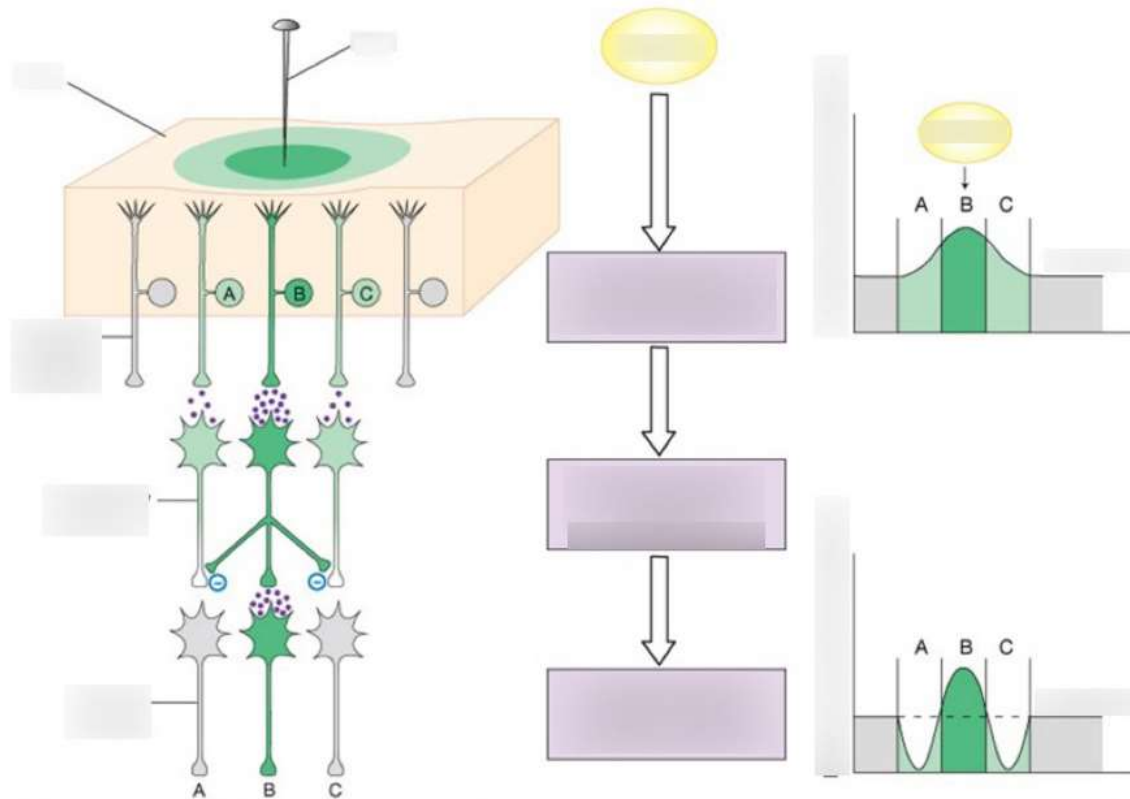
lateral inhibition

Refers to the capacity of excited neurons to reduce the activity of their neighbors.

Neurons that are firing inhibit the stimulation of surrounding neurons.

Accordingly, only the neurons that are most stimulated and least inhibited respond.

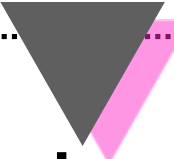
Acuity: Lateral Inhibition



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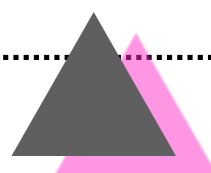
Figure 10-6: Lateral inhibition

Where is the
Stimulus?

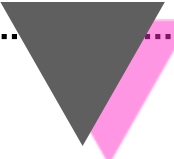


Acuity: Two Point Discrimination

1. Take your paper clip and open it up
2. Make it into a pair of forceps, the tips about 1cm apart
3. Investigate the phenomena of **two point discrimination** by placing the two points together on the fingertips, palm and forearm
4. Can you feel one or two points at these sites?



Where is the Stimulus?



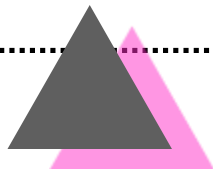
Acuity: Two Point Discrimination

- Minimal interstimulus distance required to perceive two simultaneously applied skin indentations.

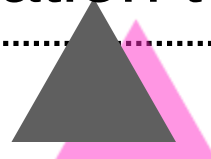
Fingertips 2 mm apart

Forearm 40 mm apart

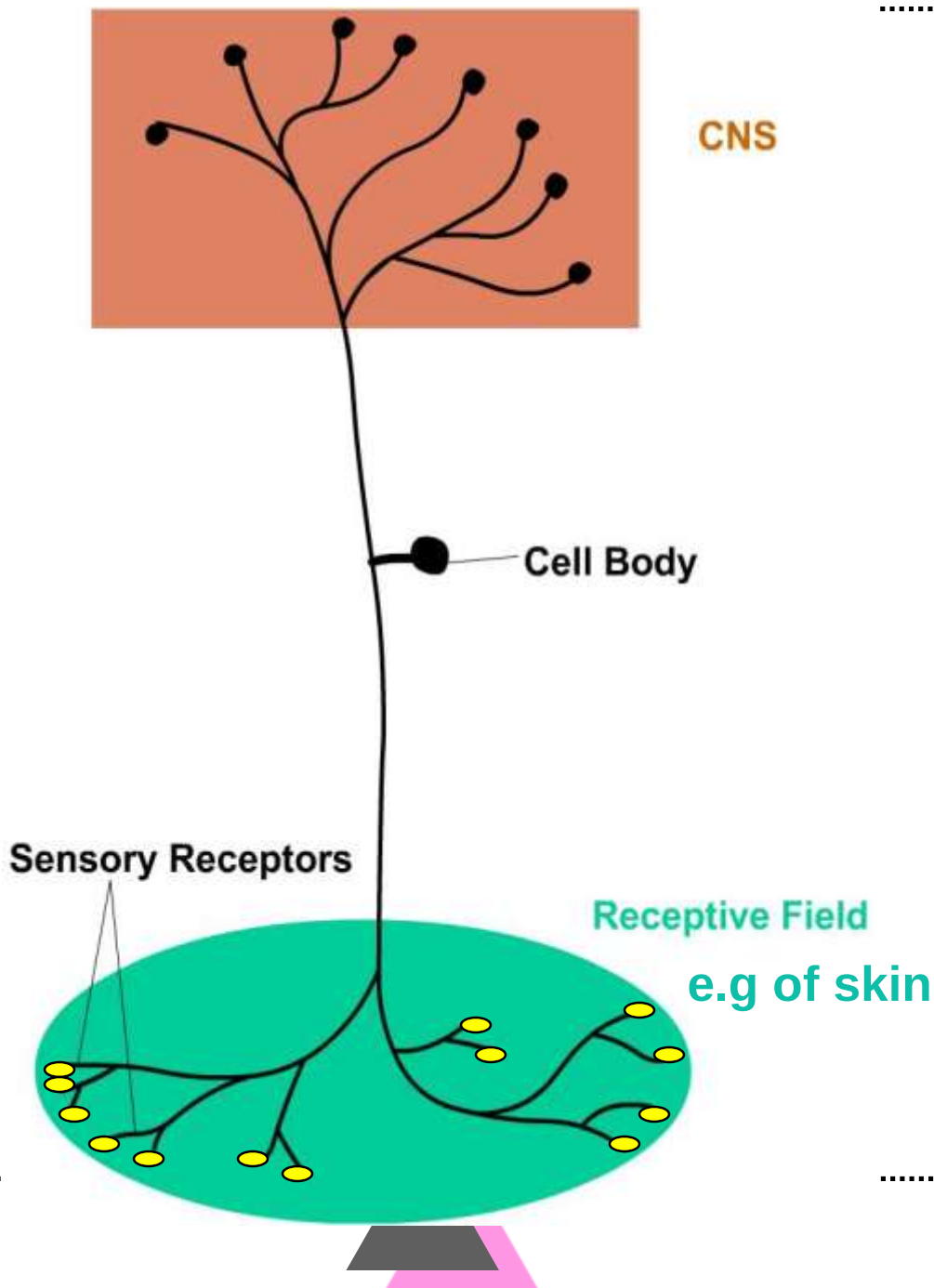
Where is the
Stimulus?



Receptive Fields



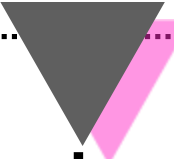
- **The receptive field** is a portion of sensory space that can elicit neuronal responses when stimulated.
- Each 1° sensory neuron picks up information from a receptive field
- Often convergence onto 2° sensory neuron $\Rightarrow$ summation of multiple stimuli
- Size of receptive field determines sensitivity to stimulus $\rightarrow$ tested by Two point discrimination test



Receptive Fields ...

- vary in size and density
- overlap with neighbouring receptive fields

Where is the Stimulus?

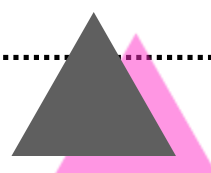


Acuity: Two Point Discrimination

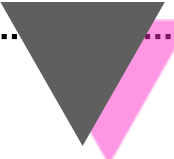
Two point discrimination is determined by:

- **Density (number) of sensory receptors**
(3-4 times greater in fingertips than other areas of hand)
- **Size of neuronal receptive fields**

Fingertips	=	1 - 2 mm
Palm	=	5 - 10 mm
- **psychological factors**



Where is the
Stimulus?

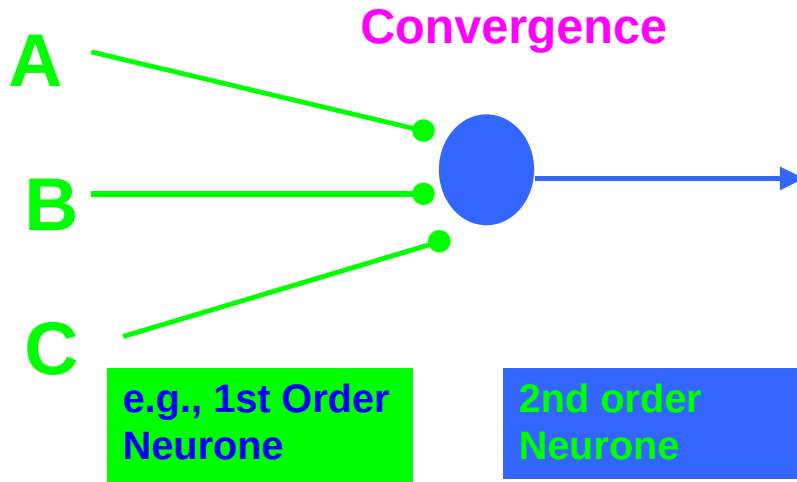


Acuity: Convergence and Divergence

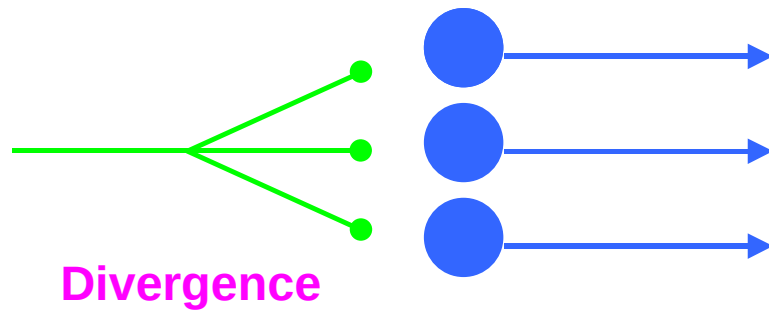
Convergence multiple sensory neuron are activated and these sensory neurons all project into (they converge on to) individual motor neurons. **So it will decrease the acuity.**

Divergence: A single sensory has multiple branches that diverge and make synaptic connections with many individual neurons. So it **will increase the acuity.**

Acuity: Convergence and Divergence



Convergence
decreases acuity



Divergence
amplifies the signal

Where is the
Stimulus?



What is somatic sensation?

How do we detect changes in our environment?

How strong is the stimulus?

How long does the stimulus last?

Where is the stimulus?

Where do we 'feel' sensation?

RIGHT SIDE
OF BODY

LEFT SIDE
OF BODY

Primary
somatosensory
area of cerebral
cortex

Thalamus

Third order
Neurone

Midbrain

Second
Order
Neurone

First
Order
Neurone

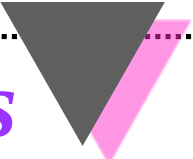
Medulla

e.g.,
touch,
vibration

Spinal Cord

**Where do we 'feel'
sensation?**

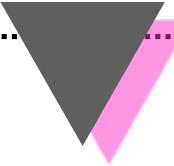
Ascending sensory pathways



The ascending sensory pathways are classified according to the functional components (modalities) they carry into:

1- The general somatic afferent (GSA) system,
which transmits sensory information such as touch, pressure, vibration, pain, temperature, stretch, and position sense from somatic structures;

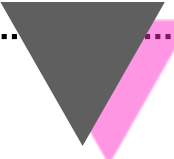
2- The general visceral afferent (GVA) system,
which transmits sensory information such as pressure, pain, and other visceral sensation from visceral structures



Ascending sensory pathways

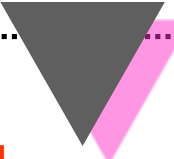
Anatomically, the ascending sensory systems consist of three distinct pathways:

- *the anterolateral system (ALS),*
- *the dorsal column–medial lemniscal (DCML) pathway,*
- *and the somatosensory pathways to the cerebellum.*



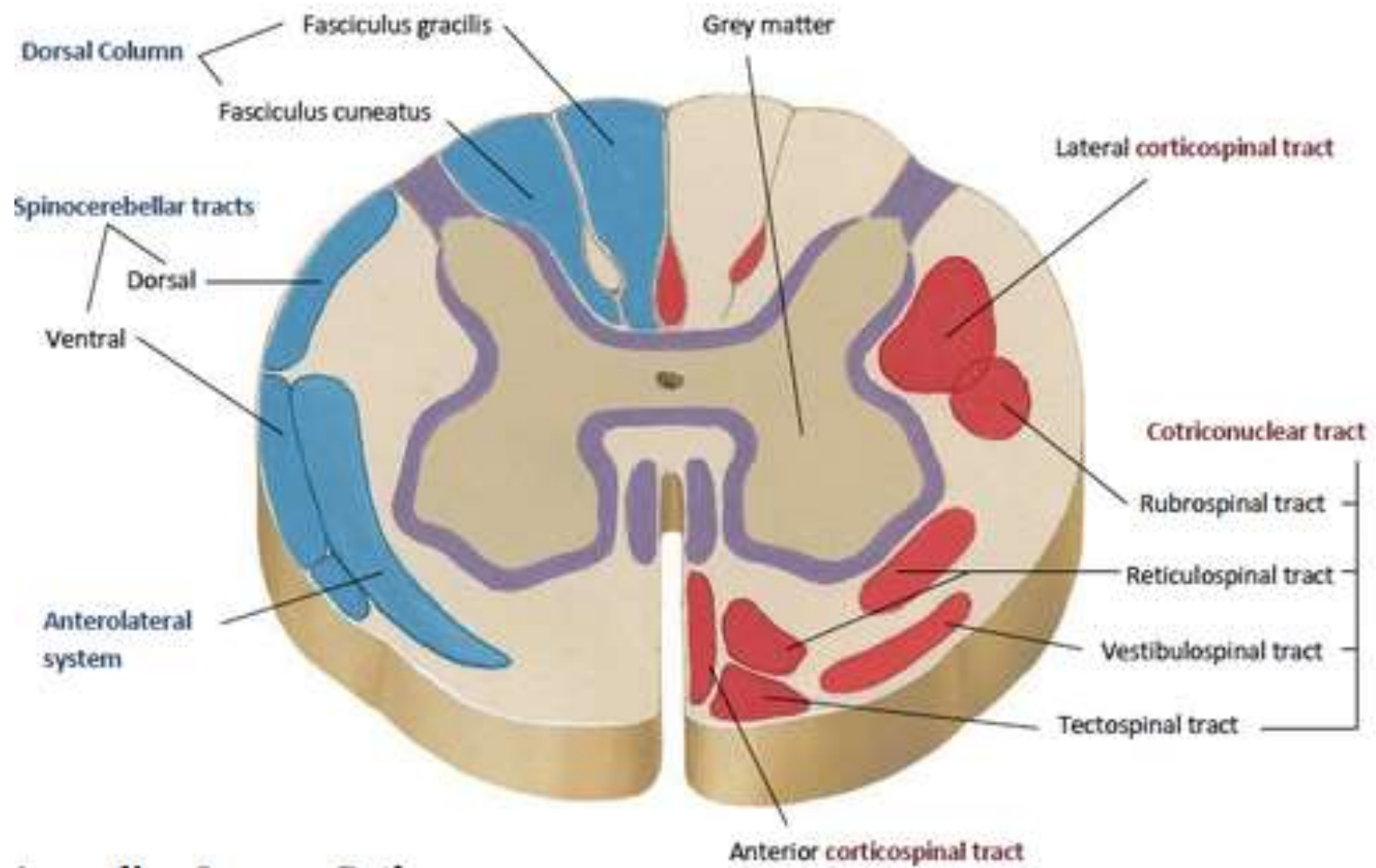
The anterolateral system

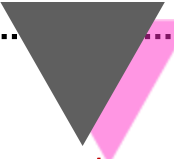
which includes the *spinothalamic*, *spinoreticular*, *spinomesencephalic*, *spinotectal*, and *spinohypothalamic tracts*, relays predominantly *pain and temperature sensation*, as well as nondiscriminative (crude or poorly localized) touch, pressure, and some proprioceptive sensation.



The dorsal column–medial lemniscal pathway

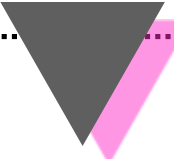
(which includes the *fasciculus gracilis*, *fasciculus cuneatus*, and *medial lemniscus*) relays discriminative (fine) tactile sense, vibratory sense, and position sense.





The somatosensory pathways to the cerebellum,

which include the *anterior, posterior, and rostral spinocerebellar*, as well as the *cuneocerebellar tracts*, relay primarily proprioceptive (but also some pain and pressure) information.

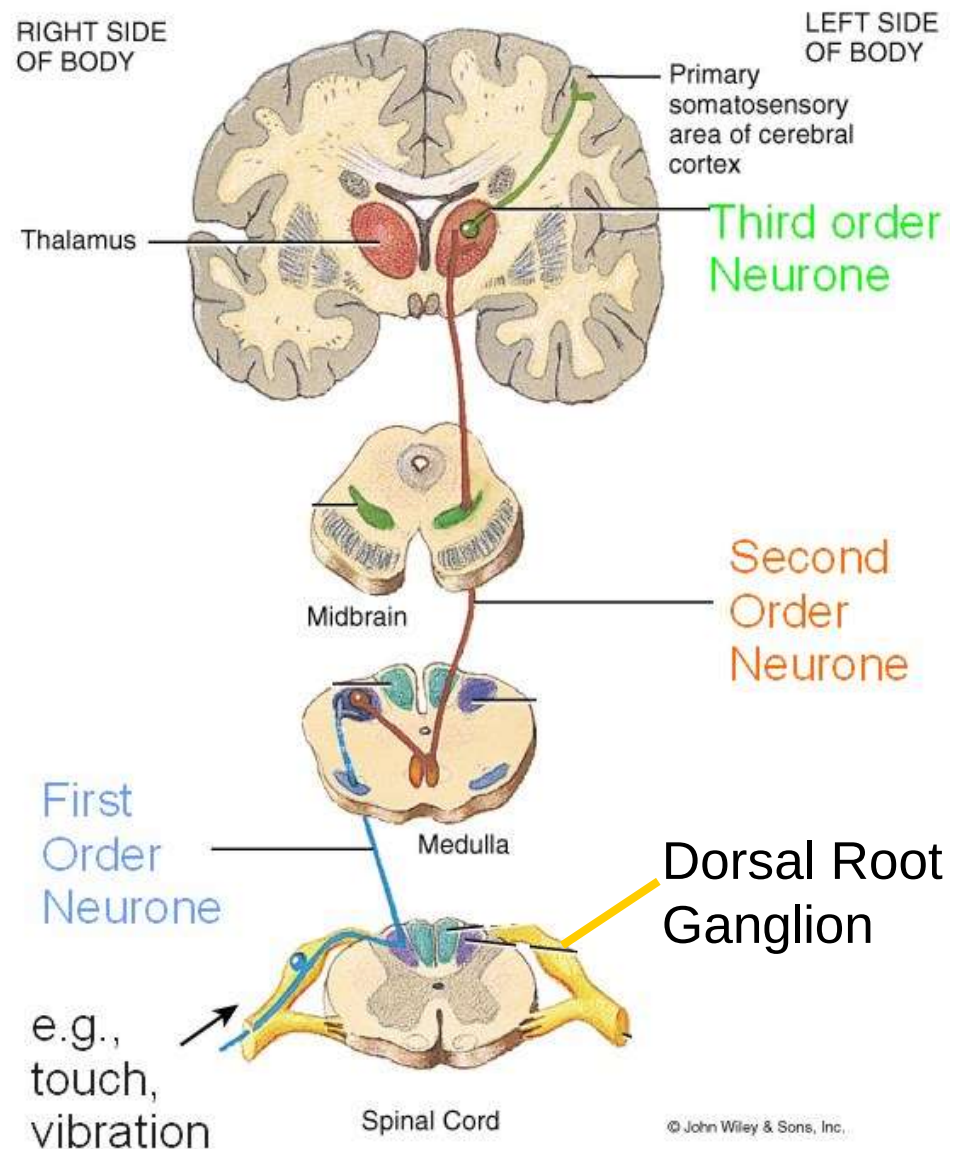


The anterolateral system (ALS)

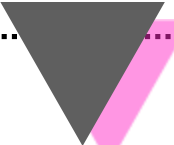
The anterolateral system (ALS) transmits **nociceptive, thermal, and non-discriminatory (crude) touch** information to higher brain centers generally by a sequence of three neurons and interneurons. The neuron sequence consists of:

Organisation of the Somatosensory System

- Third order neurone
- Second order neurone
- First order neurone
(Primary afferent neurone)

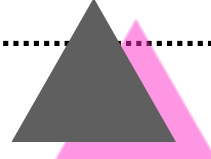


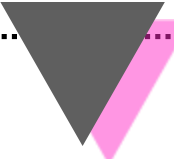
An Example



1- A first order neuron whose cell body is located in a dorsal root ganglion. It transmits sensory information from **peripheral** structures to the dorsal (posterior) horn of the spinal cord. •

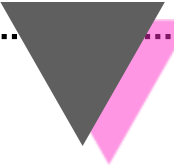
2- A second order neuron whose cell body is located within the dorsal horn of the **spinal cord**, and whose axon usually decussates and ascends to *synapse in the contralateral thalamus* (spinothalamic tract) (or called ***direct pathway***), and sending some collaterals to the reticular formation.



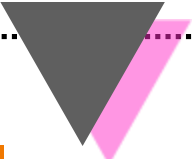


in the indirect pathway of the ALS

(spinoreticular tract) to synapse in the **reticular formation**, and sending some collaterals to the thalamus; or • as spinomesencephalic, spinotectal, or spinothalamic fibers to synapse in several brainstem nuclei.



3- A third order neuron whose cell body is located in the **thalamus**, and whose axon ascends ipsilaterally to terminate in the somatosensory cortex.



Thalamic Level

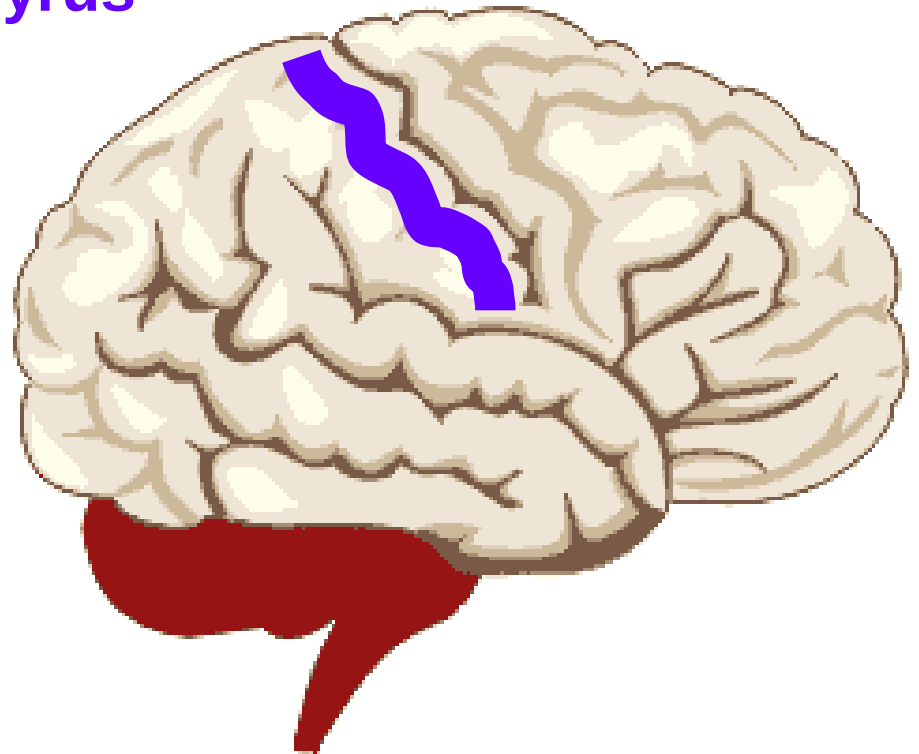
- Crude localization and discrimination of stimuli
- Highly organized projections to cortex
- Thalamic lesions: e.g., stroke, can create **thalamic overreaction**

**Where do we 'feel'
sensation?**

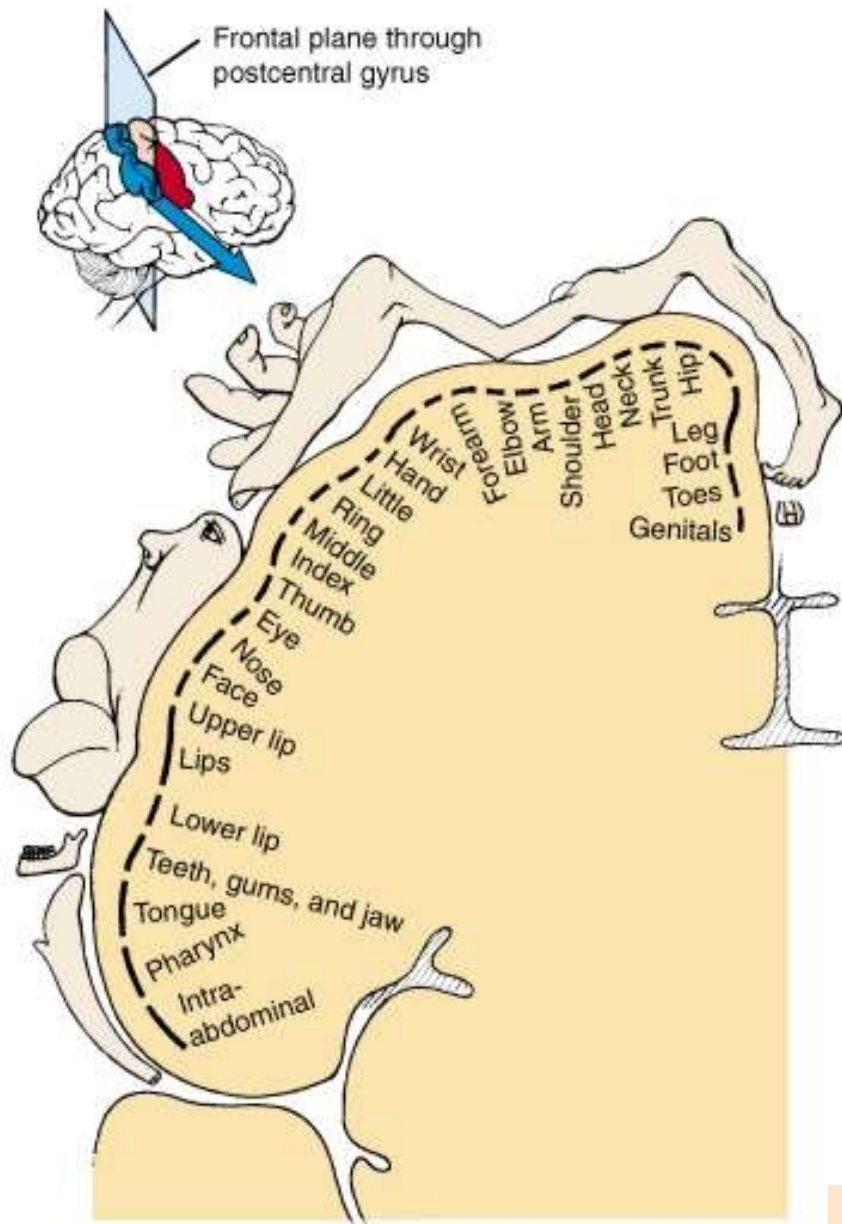
Somatosensory Cortex

- **Sharp** localisation and full recognition of qualities of modalities
- Cortical columns
- **Somatotopic representation** - each body area has specific cortical representation

Post-Central Gyrus



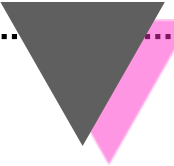
Where do we 'feel' sensation?



Somatosensory Cortex

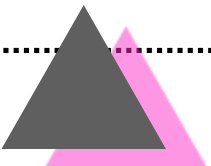
- The **sensory homunculus** (contralateral representation). The relative size of each area is reflective of the degree of sensory acuity associated with that body area

Where do we 'feel' sensation?

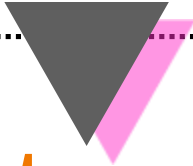


Perception

- **Perception:** is our awareness of stimuli and our ability to discriminate between different types of stimuli.

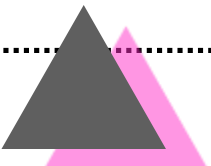


Where do we 'feel'
sensation?



Somatosensory Cortex...

- Relays to other cortical and sub-cortical areas
- Choice to respond to stimulus taken at the cortical level
- **Lesion of sensory cortex:** e.g., in repeated epileptic events, may lead to loss of two point discrimination or astereognosis



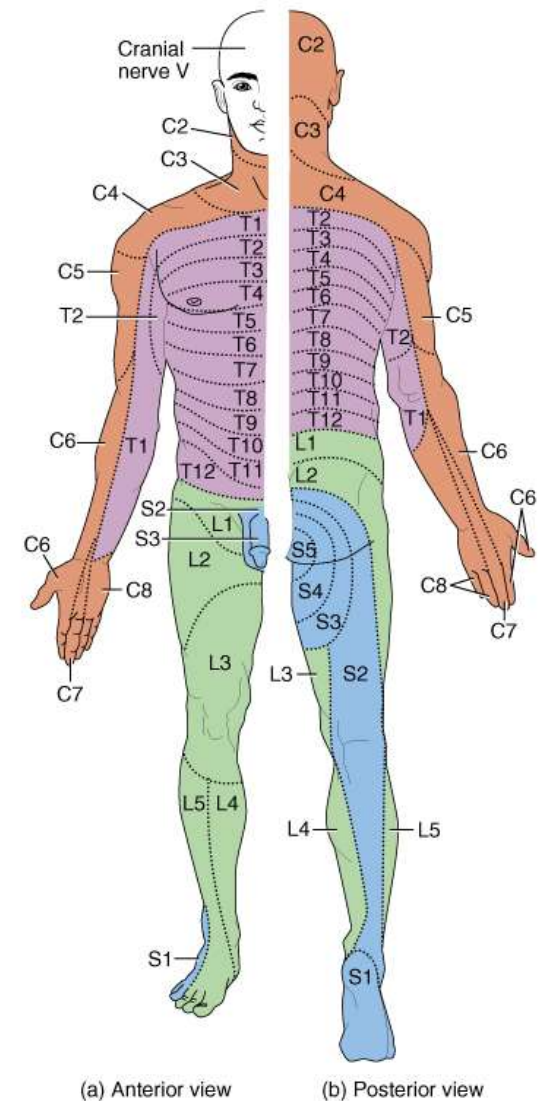
**Where do we 'feel'
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Investigation of Sensory Disturbances

- What type of sensation(s) are affected?
- Is the impairment localised?

Consider:

- peripheral nerves
- spinal nerves
- spinal tracts
- polyneuropathy



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